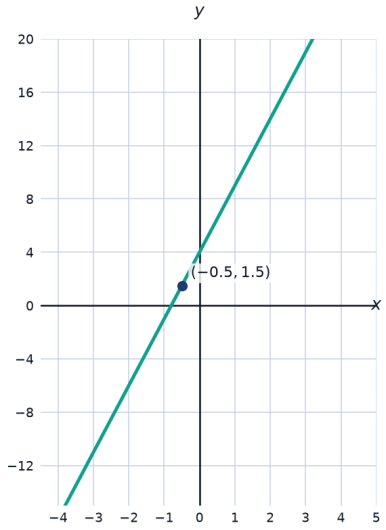
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

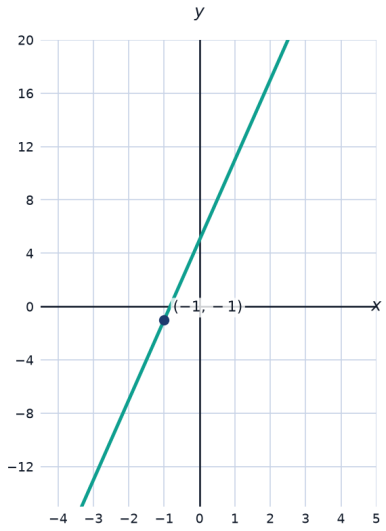
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 4$ in writing rules for relationships, evaluate $f(5)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 6x + 5$ in substitution, evaluate $f(2)$.



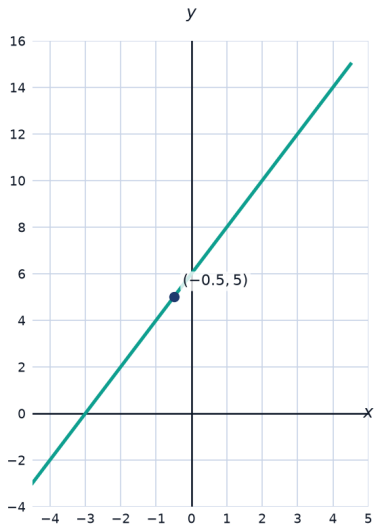
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for substitution consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean? Show your complete algebraic work.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 2x + 6$ in collecting like terms, evaluate $f(2)$.

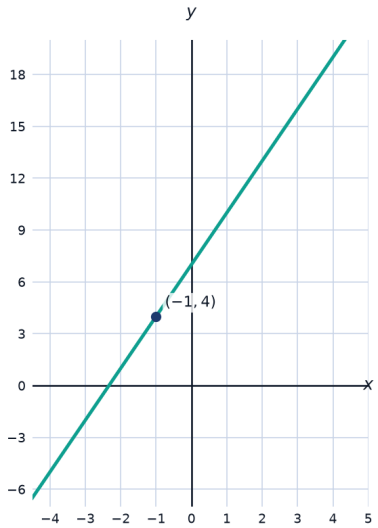


5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for collecting like terms consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean? Show your complete algebraic work.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 3x + 7$ in solving contextual problems with algebra, evaluate $f(2)$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for solving contextual problems with algebra consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean? Show your complete algebraic work.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

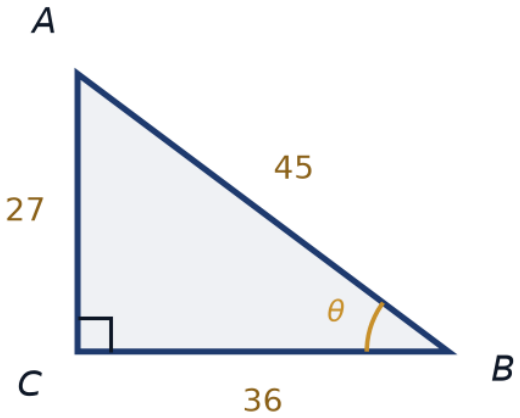
An equilateral triangle has side lengths of 12 cm. An altitude is drawn from one vertex to the opposite side. Find the exact height and area of the triangle.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In a 45°–45°–90° right triangle, the hypotenuse has length $10\sqrt{2}$ cm. What is the length of each leg, and what is the area of the triangle?

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Right Triangles & Trigonometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Trigonometry & Circular Measure · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In isosceles triangle ΔPQR, $PQ = PR$. The vertex angle $\angle P$ measures 40°. An angle bisector is drawn from base angle $\angle Q$ to point D on side PR. What is the measure of $\angle QDR$?

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the equation $\sqrt{2x + 7} = x + 2$ for all real values of x.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A circle is inscribed inside a square of side length 10 cm. A point is chosen at random inside the square. What is the exact probability that the point lands outside the circle (in the four shaded corners)?

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 2.01 Algebra & Graphs**

The function f is defined by $f(x) = x^2 - 6x + 13$ for $x \geq 3$. (a) State the range of f. (b) Find an expression for $f^{-1}(x)$ and state the domain and range of f^{-1}.

16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 3.01 Coordinate Geometry**

Functions g and h are defined by $g(x) = 2x - 1$ for $x \in \mathbb{R}$, and $h(x) = \frac{6}{x-2}$ for $x \neq 2$. (a) Find an expression for $hg(x)$ in simplest form. (b) Solve the equation $hg(x) = g(x)$.

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 4.01 Mensuration**

Find the exact area of the region enclosed between the curve $y = 4x - x^2$ and the line $y = x$.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 5.01 Geometry**

Evaluate the numerical value of $16^{-3/4} + \left(\frac{27}{8}\right)^{-2/3}$.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 7.01 Transformations & Vectors**

Two similar cylinders have surface areas of 64π cm^2 and 144π cm^2. If the volume of the smaller cylinder is 96π cm^3, find the volume of the larger cylinder.

20 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 8.01 Probability & Statistics**

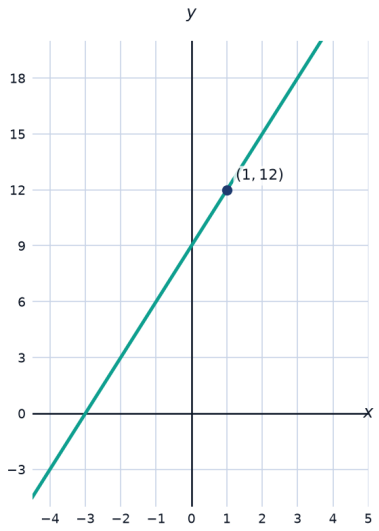
A curve has equation $y = 2x^3 - 9x^2 + 12x + 1$. (a) Find the coordinates of both stationary points. (b) Determine the nature of each stationary point using the second derivative $\frac{d^2y}{dx^2}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 9$ in introduction to variables, evaluate $f(2)$.



a. 15

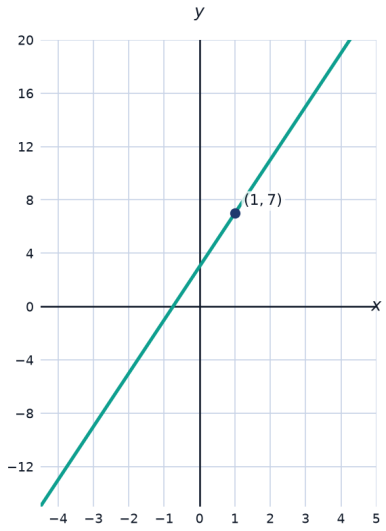
b. 18

c. 6

d. 19

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 3$ in building algebraic expressions, evaluate $f(2)$.



a. 11

b. 15

c. 8

d. 15

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for introduction to variables consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for building algebraic expressions consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

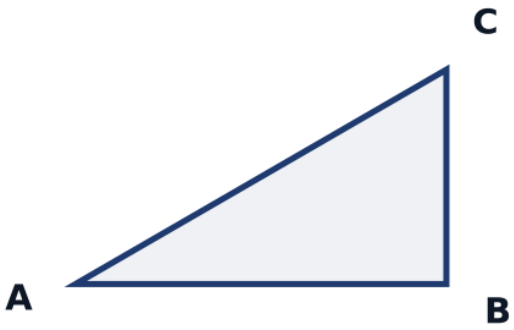
b. 21.5

c. 22.5

d. 24.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In triangle ABC, angle $A = 30^\circ$, angle $B = 90^\circ$, and side $a = 8$ cm (opposite A). Find side b (opposite B), to 2 dp.



a. 18.0 cm

b. 16.0 cm

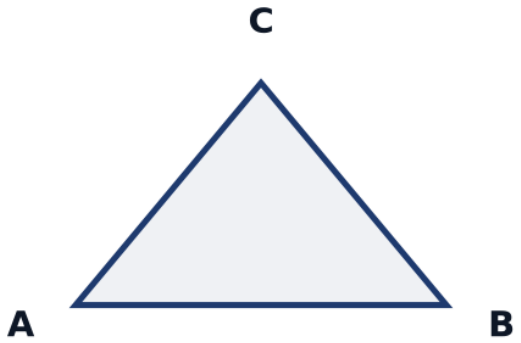
c. 17.0 cm

d. 15.0 cm

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In triangle ABC: AB=7, angle A=35°, angle B=65°. Which rule to find BC?



a. Cosine rule

b. Sine rule

c. Pythagoras

d. Area formula

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In triangle ABC, $AB = 7$ cm, $AC = 9$ cm, and angle $A = 60^\circ$. Use the cosine rule to calculate side length BC.

a. 8.19 cm

b. 8.00 cm

c. 7.50 cm

d. 8.54 cm

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In a cuboid with a square base of 8 cm by 8 cm and height 10 cm, calculate the angle that the space diagonal makes with the base (to 1 decimal place).

a. 41.4°

b. 51.3°

c. 36.9°

d. 45.0°

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In right-angled triangle ABC, angle $C = 90^\circ$, $AB = 10$ cm and angle $A = 30^\circ$. Find the length of opposite side BC.

a. 5 cm

b. 8.66 cm

c. 5.77 cm

d. 7.07 cm

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the area of triangle ABC where $a = 6$ cm, $b = 8$ cm, and included angle $C = 30^\circ$.

a. 12 cm²

b. 24 cm²

c. 20.8 cm²

d. 48 cm²

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In triangle ABC, side b = 10 cm, angle B = 45°, and angle C = 60°. Calculate the length of side c to 1 decimal place.

a. 12.2 cm

b. 8.2 cm

c. 14.1 cm

d. 10.5 cm

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In triangle PQR, side p = 5 cm, side q = 7 cm, and side r = 8 cm. Calculate angle R (to 1 decimal place).

a. 81.8°

b. 60.0°

c. 75.5°

d. 98.2°

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A ship sails 15 km on a bearing of 060° and then 20 km on a bearing of 150°. How far is the ship from its starting point?

a. 25 km

b. 35 km

c. 22.5 km

d. 30 km

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In triangle ABC, side a = 6 cm, side b = 10 cm, and side c = 14 cm. Find angle C.

a. 120°

b. 60°

c. 150°

d. 135°

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In triangle ABC, side $a = 8$ cm, angle $A = 30^\circ$, and angle $B = 45^\circ$. Find side b to 1 decimal place.

a. 11.3 cm

b. 13.8 cm

c. 9.5 cm

d. 12.0 cm

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In triangle ABC, side $b = 5$ cm, side $c = 8$ cm, and angle $A = 60^\circ$. Find side a to 1 decimal place.

a. 7.0 cm

b. 8.5 cm

c. 6.5 cm

d. 9.4 cm

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The bearing of ship B from port A is 065°. What is the back bearing of port A from ship B?

a. 115°

b. 205°

c. 245°

d. 295°

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Geometry

In right-angled triangle ABC, angle B = 90°, AB = 8 cm, and BC = 15 cm. Find the value of $\sin A$.

a. $\frac{15}{17}$

b. $\frac{8}{17}$

c. $\frac{15}{8}$

d. $\frac{8}{15}$

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Transformations & Vectors

In right-angled triangle PQR, angle Q = 90°, PQ = 5 cm, and QR = 12 cm. Find the value of $\cos P$.

a. $\frac{5}{13}$

b. $\frac{12}{13}$

c. $\frac{5}{12}$

d. $\frac{12}{5}$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Probability & Statistics

Calculate the area of triangle ABC where side $a = 6$ cm, side $b = 8$ cm, and included angle $C = 30^\circ$.

a. 12 cm²

b. 24 cm²

c. 19 cm²

d. 9 cm²

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Functions

In triangle ABC, side a = 7 cm, side b = 9 cm, and angle C = 40°. Calculate the area of triangle ABC to 1 decimal place.

a. 20.2 cm²

b. 40.5 cm²

c. 31.5 cm²

d. 24.1 cm²

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Number

The bearing of point B from point A is 045°. What is the back-bearing of A from B?

a. 225°

b. 270°

c. 315°

d. 015°

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 10.01 Sets & Venn Diagrams

In right-angled triangle XYZ, angle Y = 90°, angle X = 30°, and hypotenuse XZ = 18 cm. What is the length of side YZ?

a. 9 cm

b. 18 cm

c. $9\sqrt{3}$ cm

d. 4.5 cm

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Algebra & Graphs

From the top of a vertical cliff 40 m high, the angle of depression of a boat out at sea is 25°. How far is the boat from the base of the cliff (to 1 decimal place)?

a. 85.8 m

b. 95.2 m

c. 18.7 m

d. 44.1 m

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Coordinate Geometry

Given that $\sin x = 0.6$ where $90^\circ < x < 180^\circ$ (obtuse angle), find the value of $\cos x$.

a. -0.8

b. 0.8

c. -0.6

d. -0.4

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